# OS Skill-24

To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables.

To Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review.

nano src/main.c

```
#include <stdio.h>

int main() {

   printf("Final Project Running Successfully!\n");

   return 0;

}
```

nano Makefile

```
CC = gcc
CFLAGS = -Wall -Wextra -Werror
TARGET = project
SOURCE = src/main.c

all: $(TARGET)

$(TARGET): $(SOURCE)
	$(CC) $(CFLAGS) $(SOURCE) -o $(TARGET)

run: $(TARGET)
	./$(TARGET)

clean:
	rm -f $(TARGET)

.PHONY: all run clean
```

```
lahari@DESKTOP-AIC33F7:~$ mkdir final_project
cd final_project
mkdir src tests docs
lahari@DESKTOP-AIC33F7:~/final_project$ nano src/main.c
lahari@DESKTOP-AIC33F7:~/final_project$ nano Makefile
lahari@DESKTOP-AIC33F7:~/final_project$ make
gcc -Wall -Wextra -Werror src/main.c -o project
lahari@DESKTOP-AIC33F7:~/final_project$ make run
./project
Final Project Running Successfully!
lahari@DESKTOP-AIC33F7:~/final_project$ make clean
rm -f project
lahari@DESKTOP-AIC33F7:~/final_project$ nano README.md
lahari@DESKTOP-AIC33F7:~/final_project$ find . -maxdepth 2 -type f
./Makefile
./README.md
./src/main.c
lahari@DESKTOP-AIC33F7:~/final_project$ ./src/main.c
./Makefile
./README.md
-bash: ./src/main.c: Permission denied
-bash: ./Makefile: Permission denied
-bash: ./README.md: Permission denied
lahari@DESKTOP-AIC33F7:~/final_project$ find . -maxdepth 2 -print
.
./Makefile
./docs
./README.md
./src
./src/main.c
./tests
lahari@DESKTOP-AIC33F7:~/final_project$ git --version
git version 2.43.0
```

```
lahari@DESKTOP-AIC33F7:~/final_project$ git init
git add src/main.c Makefile README.md
git commit -m "Initial project release"
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/lahari/final_project/.git/
[master (root-commit) c1343ad] Initial project release
 3 files changed, 62 insertions(+)
 create mode 100644 Makefile
 create mode 100644 README.md
 create mode 100644 src/main.c
lahari@DESKTOP-AIC33F7:~/final_project$ git config --global user.name "Your Name"
git config --global user.email "you@example.com"
lahari@DESKTOP-AIC33F7:~/final_project$ git tag -a v1.0.0 -m "First stable release"
lahari@DESKTOP-AIC33F7:~/final_project$ git tag
git log -1 --oneline
v1.0.0
c1343ad (HEAD -> master, tag: v1.0.0) Initial project release
lahari@DESKTOP-AIC33F7:~/final_project$ make clean
make
make run
git status
git tag
rm -f project
gcc -Wall -Wextra -Werror src/main.c -o project
./project
Final Project Running Successfully!
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        project

nothing added to commit but untracked files present (use "git add" to track)
v1.0.0
lahari@DESKTOP-AIC33F7:~/final_project$
```

nano demo.c

```
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handle_signal(int sig) {
   printf("\nReceived signal %d. Exiting safely.\n", sig);
   fflush(stdout);
   _exit(0);
}
```

```
int main() {
    signal(SIGINT, handle_signal);

    printf("Demo program started.\n");
    printf("Press Ctrl+C to test signal handling.\n");
    fflush(stdout);

    while (1) {
        pause();
    }

    return 0;
}
```

```
lahari@DESKTOP-AIC33F7:~/final_project/demo_lab$ nano demo.c
lahari@DESKTOP-AIC33F7:~/final_project/demo_lab$ gcc -Wall -Wextra demo.c -o demo
lahari@DESKTOP-AIC33F7:~/final_project/demo_lab$ ./demo
Demo program started.
Press Ctrl+C to test signal handling.
^C
Received signal 2. Exiting safely.
lahari@DESKTOP-AIC33F7:~/final_project/demo_lab$ seq 1 100 | grep '5' | sort -n | head -10
5
15
25
35
45
50
51
52
53
54
lahari@DESKTOP-AIC33F7:~/final_project/demo_lab$ seq 1 100 | grep '5' | sort -n | head -10 > pipeline_output.txt
lahari@DESKTOP-AIC33F7:~/final_project/demo_lab$ cat pipeline_output.txt
5
15
25
35
45
50
51
52
53
54
lahari@DESKTOP-AIC33F7:~/final_project/demo_lab$ sleep 300 &
[1] 806
lahari@DESKTOP-AIC33F7:~/final_project/demo_lab$ echo $!
806
lahari@DESKTOP-AIC33F7:~/final_project/demo_lab$ sleep 300 &
PID=$!
echo "Process ID: $PID"
[2] 807
Process ID: 807
lahari@DESKTOP-AIC33F7:~/final_project/demo_lab$ kill -TERM "$PID"
lahari@DESKTOP-AIC33F7:~/final_project/demo_lab$ ps -p "$PID"
    PID TTY          TIME CMD
[2]+  Terminated              sleep 300
lahari@DESKTOP-AIC33F7:~/final_project/demo_lab$
```